

DECODED EXPLAINED

30th August, 2026

FSSAI's Proposed Front-of-Pack Warning Labels

Soon, red hexagonal label to warn of high fat, sugar, salt in snacks

Ananthakrishnan G
& Anonna Dutt
New Delhi, August 29

PROMPTED BY the Supreme Court, the Food Safety and Standards Authority of India (FSSAI) has proposed introducing a red hexagonal front-of-pack label to warn consumers if packaged foods are high in salt, added sugar, or added fat.

"The proposal is intended to provide a simple, prominent and easily comprehensible warning to consumers regarding food products which are high in specified nutrients of concern," the compliance affidavit filed by Kavitha Ramaswamy, Joint Director, FSSAI, before the SC said. The apex court is hearing a plea by a Kerala-based non-profit organisation, 3S and Our Health Society.

While there is no notification yet — nor a timeline for implementation — the country's apex food safety regulator has stated

in its affidavit that the warning label, in English, will be carried on the front of the packet in a font size one point larger than the one used in the nutritional information table provided at the back of the pack.

The plan to introduce the Front-of-Pack Nutrition Labeling will be executed in two phases, as per the affidavit. In the first phase, only products with a high content of at least two of the ingredients will carry the warning label. In the second phase, the warning will extend to products high in even one ingredient.

The FSSAI affidavit proposes the warning label for packaged foods high in two or more ingredients of concern based on thresholds specified under the Dietary Guidelines for Indians, 2024, issued by the Indian Council of Medical Research-National Institute of Nutrition.

"The label shall indicate the applicable declarations such as 'high fat', 'high sugar', 'high salt'

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Nutrient threshold, use of 'added' term

EXPERTS SAY the two or more nutrient threshold in phase I may render the move ineffective as many unhealthy products would not need to add warning label since they are high in just one nutrient. They also flagged the challenge of the use of the term "added" as it is difficult to estimate.

and/or 'highly sweetened beverage', as the case may be, to enable consumers to identify products high in the specified nutrients," the affidavit states.

Single-ingredient food products and "food products which are inherently rich in fat, sugar

or salt, such as ghee, edible oil, salt, sugar, jaggery and honey, subject to the applicability of other requirements under the food safety and labelling regulations", will be exempted from the labelling requirement, the affidavit stated. The FSSAI said the two-phase plan is "to facilitate consumer acceptability and provide industry adequate time for reformulation".

Advocate Rajiv Shankar Divedi, appearing for the petitioner, said, "We have received the FSSAI compliance affidavit. We have prepared a revised draft, which has been circulated on the petitioner's website. We have invited comments from the public on the said rejoinder (affidavit). After getting all the comments, we will file our rejoinder."

Nutrition experts have welcomed the move to introduce warning labels but have some concerns about the implementation. Dr Arun Gupta, Convener of Nutrition Advocacy in Public

Interest (NAPI)- India, said: "In principle, FSSAI has agreed to the warning labels, but they have put in a whole lot of conditions that make it ineffective. Take for example the two or more nutrient threshold in the phase I of implementation. Many unhealthy products would not need to add the warning label because they are high in just one nutrient." He said the FSSAI has also not given any timeline for phase II when all products high in fats, sugars or salts will have to carry the warning.

Another challenge is the use of the term "added". "The food products will have to carry the warning only when there are added sugars or added fats. One, this is difficult to estimate. Two, it is not just the added fat or added sugar that harms the health; high levels of natural sugars can also be bad." He also said the font size proposed is too small and the message should be carried in Hindi as well.

'FSSAI's proposed warning labels could let unhealthy foods escape scrutiny'

Bindu Shajan Perappadan
NEW DELHI

The Food Safety and Standards Authority of India (FSSAI) has proposed red hexagonal warning labels on packaged foods high in sugar, salt, and saturated fat, but critical gaps in the framework could leave several unhealthy products with no warning, nutrition advocate Arun Gupta said.

While the proposal marks a step forward after sustained intervention by the Supreme Court, its "two or more" nutrients threshold is a major loophole, Dr. Gupta, convener of Nutrition Advocacy in Public Interest (NAPI), told *The Hindu*.

The Supreme Court recently pushed the Centre and FSSAI to act swiftly on front-of-pack warning labels for unhealthy packaged foods. The FSSAI has now proposed red, hexagonal labels flagging products high in sugar, salt or fat to help consumers make informed choices.

Faulty framework

Under the proposal, a warning would be triggered only when a product is high in two or more nutrients of concern — added saturated fat, added sugar, and added salt. "A product



The top court recently pushed the Centre and the FSSAI to act swiftly on front-of-pack warning labels for packaged foods.

could be high in sugar but carry no warning if it is low in fat and salt. That defeats the purpose of a warning label," Dr. Gupta said.

The rule, he said, should be simple: "High in sugar — warning; high in salt — warning; high in saturated fat — warning." He also called for time-bound implementation, and said the framework should be aligned with the Indian Council of Medical Research-National Institute of Nutrition (ICMR-NIN) Dietary Guidelines, 2024, which provide thresholds for total sugar and total fat per 100 gm.

Dr. Gupta also questioned the proposed use of "added" nutrients and called for following a clear, scientifically robust definition of high fat, sugar and salt (HFSS) foods. "India already has an HFSS definition reflected in the 2022 draft notification and par-

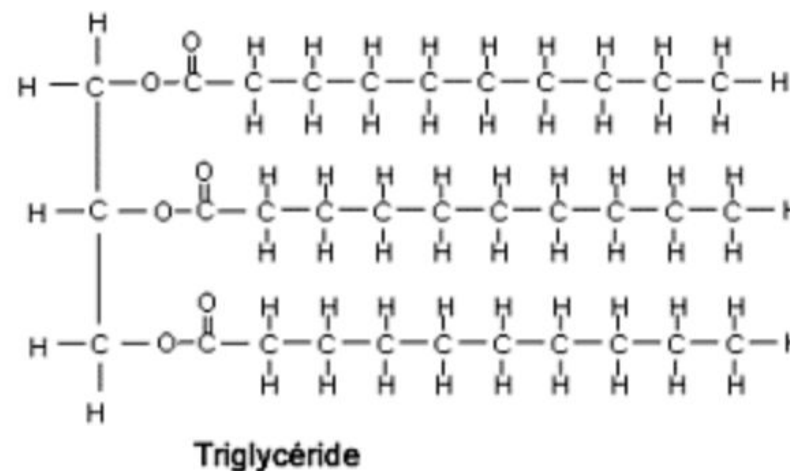
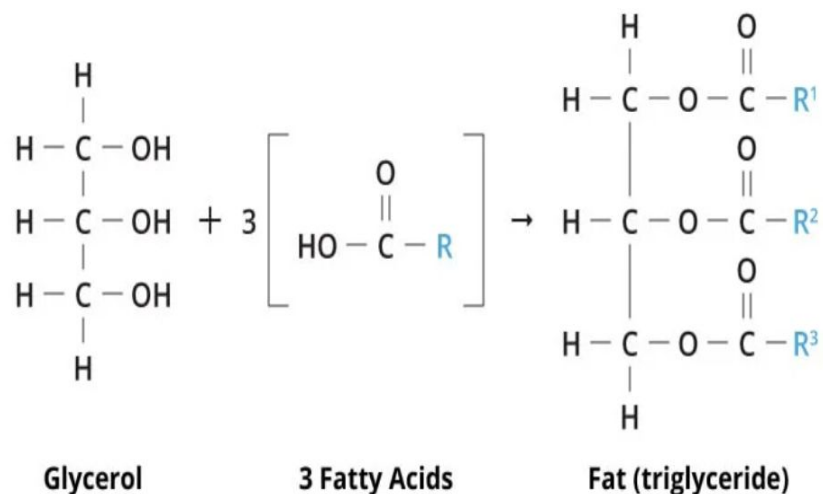
liamentary discussions. We should not keep reinventing the definition in a way that weakens the regulation," he said.

He also criticised the proposed size of the warning, which would be one point larger than the back-of-pack nutrition information. "If the nutrition table is printed in tiny type, making the warning one point larger does not make it prominent," he said, calling for the size of the warning to be linked to the principal display area of the package.

India has mandatory nutrition information on packaged foods but has never had a mandatory front-of-pack warning specifically flagging products high in sugar, salt or saturated fat. FSSAI proposed HFSS and front-of-pack nutrition regulations in 2022, but the process has remained stalled.

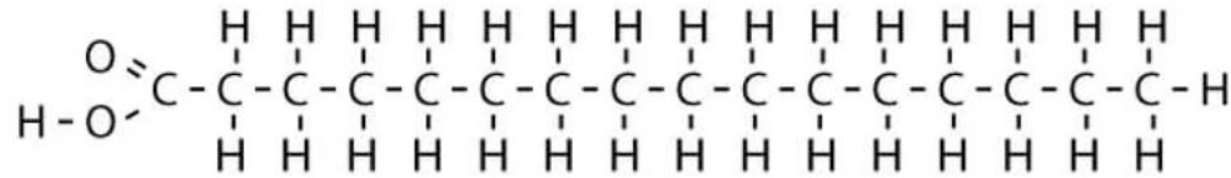
FSSAI's Proposed Front-of-Pack Warning Labels

- **Fat is one of the three main macronutrients: fat, carbohydrate, & protein.**
- **Fat is a major source of energy & helps your body absorb vitamins**
- **Fats, also known as triglycerides, are esters of three fatty acid chains & the alcohol glycerol.**

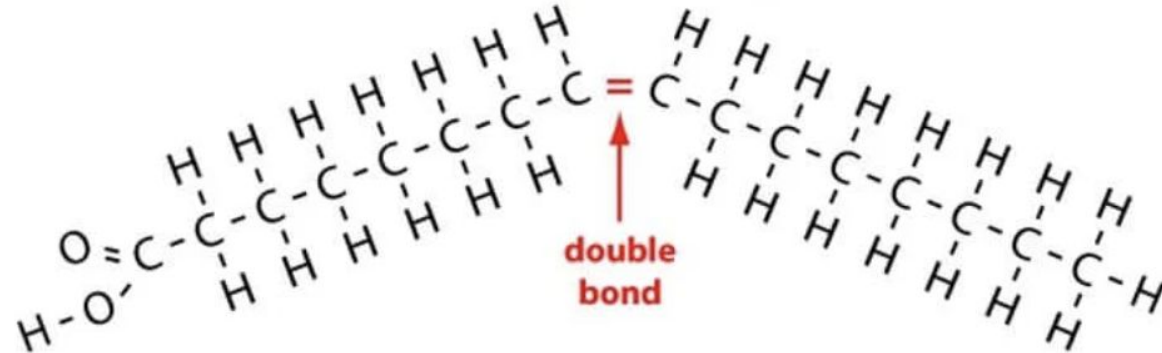


FSSAI's Proposed Front-of-Pack Warning Labels

saturated fatty acid



unsaturated fatty acid



FSSAI's Proposed Front-of-Pack Warning Labels

Saturated fats

A saturated fat is a fat in which the fatty acids all have single bonds.

A saturated fat has the maximum number of hydrogens bonded to the carbons, & therefore is 'saturated' with hydrogen atoms.

Most animal fats are saturated whereas the fats of plants & fish are generally unsaturated.

Many experts recommend a diet low in saturated fat.

Too much saturated fat can cause cholesterol to build up in your arteries (blood vessels). Saturated fats raise your LDL (bad) cholesterol. High LDL cholesterol increases your risk for heart disease and stroke.

Saturated fats are popular with manufacturers of processed foods because they are less vulnerable to rancidity & are, in general, more solid at room temperature than unsaturated fats.

Unsaturated fats

An unsaturated fat is a fatty acid in which there is at least one double bond within the fatty acid chain.

Where double bonds are formed, hydrogen atoms are eliminated.

Unsaturated fats are healthy fats. They can be broken into two categories: monounsaturated fats and polyunsaturated fats.

These fats can actually help lower your cholesterol,

The greater the degree of unsaturation in a fatty acid (i.e., the more double bonds in the fatty acid) the more vulnerable it is to rancidity (lipid oxidation or rusting of fats).

Antioxidants can protect unsaturated fat from lipid oxidation.

FSSAI's Proposed Front-of-Pack Warning Labels

Trans Fats

Trans fats or trans-unsaturated fatty acids are a type of unsaturated fats that are uncommon in nature.

Trans fat is simply liquid oils turned into solid fats during food processing.

Natural trans fats, present in very small amounts in certain animal products, are not considered harmful.

Foods containing trans fat are usually labelled as “partially hydrogenated”.

Partially hydrogenated oil is less likely to spoil, so foods made with it have a longer shelf life.

Trans fats are worse than saturated fats.

Trans fats are easy to use, inexpensive to produce & last a long time.

Trans fats give foods a desirable taste & texture.

Regulation

FSSAI had capped trans fats in oils & fats to 3% by 2021, & 2% by 2022 from the current levels of 5%.

From 01/01/2022, India will limit trans fat to 2% by mass of the total oils/fats present in a food product.

WHO has called for the elimination of industrially-produced trans-fatty acids from the global food supply by 2023.

REPLACE action framework by WHO (2018): Provides a strategic approach to eliminating industrially produced trans fat from national food supplies.

FSSAI's Proposed Front-of-Pack Warning Labels

FSSAI: Front-of-Pack Warning Label

Proposal: FSSAI has proposed a red hexagonal warning label on packaged foods high in added sugar, salt or saturated fat, following Supreme Court intervention.

India currently mandates nutrition information on packaged foods but has no mandatory front-of-pack warning for these nutrients. FSSAI first proposed such regulations in 2022.

Phase I: Warning applies only when a product is high in at least 2 of 3 nutrients:

Added sugar

Added salt

Added saturated fat

Phase II: Warning would apply when a product is high in even one nutrient. Timeline not specified.

Basis: Thresholds are based on the ICMR-NIN Dietary Guidelines for Indians, 2024.

Key criticism : “Two-or-more” loophole: A product high in sugar alone could escape the warning if it is low in salt and saturated fat.

FSSAI's Proposed Front-of-Pack Warning Labels

The Food Safety and Standards Authority of India (FSSAI) has been established under the Food Safety and Standards Act, 2006, which consolidates various acts and orders that had hitherto handled food-related issues across various Ministries and Departments.

FSSAI has been created for laying down science-based standards for articles of food and to regulate their manufacture, storage, distribution, sale, and import, to ensure the availability of safe and wholesome food for human consumption.

Various central Acts such as the Prevention of Food Adulteration Act, 1954; Fruit Products Order, 1955; Meat Food Products Order, 1973; Vegetable Oil Products (Control) Order, 1947; Edible Oils Packaging (Regulation) Order, 1988; Solvent Extracted Oil, De-Oiled Meal and Edible Flour (Control) Order, 1967; and Milk and Milk Products Order, 1992 were repealed after the commencement of the FSS Act, 2006.

The Act also aims to establish a single reference point for all matters relating to food safety and standards, by moving from a multi-level, multi-departmental control system to a single line of command. To this effect, the Act establishes an independent statutory authority the Food Safety and Standards Authority of India with its head office at Delhi. FSSAI and the State Food Safety Authorities enforce the various provisions of the Act.

FSSAI's Proposed Front-of-Pack Warning Labels

The Ministry of Health & Family Welfare, Government of India, is the administrative ministry for the implementation of FSSAI. The Chairperson and Chief Executive Officer of FSSAI are appointed by the Government of India, with the Chairperson holding the rank of Secretary to the Government of India.

FSSAI's Proposed Front-of-Pack Warning Labels

A company marketing food products advertises that its items do not contain trans-fats. What does this campaign signify to the customers?

- 1. The food products are not made out of hydrogenated oils.**
- 2. The food products are not made out of animal fats/oils.**
- 3. The oils used are not likely to damage the cardiovascular health of the consumers.**

Which of the statements given below is/are correct?

This question was previously asked in

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1. 1 only

2. 2 and 3 only

3. 1 and 3 only

4. 1, 2 and 3

FSSAI's Proposed Front-of-Pack Warning Labels

Consider the following statement:

1. The Food Safety and Standards Act, 2006 replaced the Prevention of Food Adulteration Act, 1954.
2. The Food Safety and Standards Authority of India (FSSAI) is under the charge of Director General of Health Services in the Union Ministry of Health and Family Welfare.

Which of the statements given above is/are correct?

This question was previously asked in
Official UPSC Civil Services Exam
2018 Prelims Part A

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1. 1 only

2. 2 only

3. Both 1 and 2

4. Neither 1 nor 2

A Quiet Move That Has Kept Dehradun's Vultures Safe

A quiet move that has kept Dehradun's vultures safe

Rahul Karmakar
GUWAHATI

Poisoning, veterinary non-steroidal anti-inflammatory drugs (NSAIDs), food scarcity, and habitat loss continue to remain survival challenges for vultures in India.

A decade-old study, published in the latest issue of the *Journal of Threatened Taxa*, suggests that relocating a livestock carcass dump 2.4 km away from high-tension power infrastructure near Uttarakhand's Dehradun may have saved these large scavengers from another existential threat – electrocution.

The authors of the study are Javed Anver, Yash Vardhan Singh Sengar, Bikash

Ghimire, Vonchano Ngulie, Hemam Rama Nanda, Tashi P. Dorji, Ankit Sharma, Akshay Jain, Sas Biswas, Arun Kumar, and M. Muzahed.

Vultures are among the most threatened birds globally, having undergone severe population declines across Asia and Africa due to multiple anthropogenic pressures. According to the study, electrocution from power infrastructure has been the most under-reported threat.

The study documented electrocution mortality of vultures in the Sudhauwala region of Dehradun, based on 34 surveys conducted between May 2011 and February 2014. The study recorded 743 individual vul-



Vultures are among the most threatened birds globally, having undergone severe population declines. SPECIAL ARRANGEMENT

tures belonging to five species.

“During the study period, 46 vulture carcasses attributable to electrocution were documented in proximity to high-tension

power transmission structures. The highest proportion of mortality relative to observations was recorded for the Himalayan griffon (*Gyps himalayensis*), followed by the Eurasian grif-

fon (*Gyps fulvus*), and the cinereous vulture (*Aegypius monachus*),” the study said.

“Although surveys were conducted on a non-systematic basis, the number of mortality events observed suggests that electrocution may represent a significant localised threat, particularly for long-lived scavengers such as vultures,” it said.

Reduced mortality

Based on these observations, local authorities relocated the carcass dumping site about 2.4 km away from the high-risk electricity infrastructure in April 2015. Subsequent observations indicated continued use of the new site by vultures, with no mortality

events recorded during limited post-relocation monitoring.

Other raptor species, including the black kite (*Milvus migrans*) and crested serpent eagle (*Spilornis cheela*), were also recorded at the relocated site.

“Our findings indicated that relocating carcass dumping sites away from hazardous power infrastructure can be a practical conservation measure for reducing electrocution risk and could potentially be applied in other affected areas,” Mr. Anver of the Dehradun-based Wildlife Institute of India told *The Hindu*.

However, no studies have been conducted after 2015.

A Quiet Move That Has Kept Dehradun's Vultures Safe

- **Vultures are sociable creatures and are often seen as a collective unit. Out of 22 species of vultures in the world, nine are found in India.**
- **IUCN**
 1. **White rumped vulture (Critically Endangered)**
 2. **Slender billed vulture (Critically Endangered)**
 3. **Long billed vulture (Critically Endangered)**
 4. **Red headed vulture (Critically Endangered)**
 5. **Egyptian vulture (Endangered)**
 6. **Himalayan Griffon (Near Threatened)**
 7. **Cinereous vulture (Near Threatened)**
 8. **Bearded vulture (Near Threatened)**
 9. **Griffon Vulture (Least Concern).**

A Quiet Move That Has Kept Dehradun's Vultures Safe

- **Role of Vultures**

- **Carcass removal and sanitation** : If dead animals are left to rot for long durations, it may give rise to disease-causing pathogens.
- **Nutrient cycling and food-web function**
- **Regulation of the scavenger community**

- **Issues**

- **They are slow breeders**
- **Use of diclofenac, ketoprofen and aceclofenac (NSAIDS)** : The main reason for the decline in the vulture population is the use of the drug, diclofenac. Diclofenac, which relieves cattle of pain, is toxic to vultures even in small doses and causes kidney failure and death. (Meloxicam can be used as an alternative)
- **Hunting**
- **Quarrying**
- **Food scarcity**
- **Electrolution**

- **Measures**

- **India banned diclofenac for veterinary use in 2006.**
- **It banned the manufacture, sale, and distribution of ketoprofen and aceclofenac, which are other non-steroidal anti-inflammatory drugs (NSAIDs) toxic to vultures. (2023)**
- **Vulture 'restaurants', which exist in some countries, are also a way of preserving the population.**

A Quiet Move That Has Kept Dehradun's Vultures Safe

Vultures which used to be very common in Indian countryside some years ago are rarely seen nowadays. This is attributed to

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1. the destruction of their nesting sites by new invasive species.
2. a drug used by cattle owners for treating their diseased cattle.
3. scarcity of food available to them.
4. a widespread, persistent and fatal disease among them .

Let the Western Ghats breathe



ASEEM SHRIVASTAVA

THE CANOPY closes overhead as the monsoon sky lowers, and the Katlekan dark forest of North Kanara slips into an almost absolute blackness. It is the middle of the day. A few hundred yards in, the light dies so completely that you reach for a torch just to discern the brook gurgling a few yards away. Hornbills call through the dripping understorey, mountain imperial pigeons murmur in the vaulted gloom, and eagle owls send low hoots echoing between ancient trunks. Here, in this living darkness, the dry, fragmented woodlots that pass for forests in “New India” reveal themselves as paltry shadows of the *aranya* that inspired poets like Rabindranath Tagore — a primordial realm that still breathes and governs itself, reminding all that true wilderness does not merely survive in India; it thrives in the damp air and the dark. If we let it.

Having done my doctoral dissertation on forest management and seen jungles across four continents, I thought I knew what a forest was. I was wrong.

The Katlekan darkness is merely the most intimate face of a mountain chain whose influence extends far beyond its forested slopes. The Western Ghats, running 1,600 kilometres along the edge of the pen-

insula, are nothing less than the living ecological spine of India. Recognised as one of the world’s eight hottest biodiversity hot-spots, these mountains harbour extraordinary life — nearly half of India’s flowering plant species, a third of its vertebrate fauna, and over 300 globally threatened species find refuge here. Several species, like the Katlekan Marsh Nut and the Gund Day Gecko, are exclusive to the very forests I visited.

Yet the significance of the Ghats runs deeper than species counts. They gather the first impact of the southwest monsoon and release it as the lifeblood of the Deccan: The Godavari, Krishna, Kaveri, and Tungabhadra rivers originate here, sustaining hundreds of millions across six states. Their dense forests temper regional climate, draw down carbon, hold the soil in place, and soften the runoff and flooding that would otherwise devastate downstream plains.

It is no exaggeration, then, to call these rainforests “India’s Amazon”. Like their South American counterpart, they behave as a vast living organism — breathing in carbon dioxide, exhaling oxygen, and cycling moisture back into the atmosphere to sustain the monsoon itself. Their layered canopies, dripping with epiphytes and echoed by the calls of hornbills and lion-tailed macaques, weave a self-renewing tapestry of life that sustains the climate and waters of the entire peninsula. Just as the Amazon Basin constitutes the lungs of South America, the Western Ghats are the lungs and wellspring of India. As ecologists like Madhav Gadgil have long noted, their continued integrity is not merely a conservation priority; it is a matter of civilisational ecological survival.

The Katlekan darkness is merely the most intimate face of a mountain chain whose influence extends far beyond its forested slopes

In pursuit of ‘development’

Meanwhile, something else is planned in Katlekan’s neighbourhood. The proposed Rs 10,000-crore, 2,000 MW Sharavathi Pumped Storage Project in Karnataka’s Shimoga and Uttara Kannada district remains under an interim stay by the Karnataka High Court, with work frozen until further orders. It aims to link the Talakalale and Gerusoppa reservoirs on the heavily hydropower-dammed Sharavathi, through massive underground tunnels and powerhouses, pumping water uphill during off-peak hours and releasing it to generate power during peak demand.

The project sits entirely within the Sharavathi Valley Lion-Tailed Macaque Wildlife Sanctuary, which enjoys one of the highest levels of protection under the Wildlife Act, where such infrastructure is strictly prohibited. An expert panel constituted by a Standing Committee of National Board of Wildlife recommended against approval, citing irreversible ecological harm and questioning its technical justification. The panel found it would yield no new net energy to the grid, instead producing a net loss of approximately 500 MW due to pumping requirements.

The stakes are enormous. The project requires clearing over 50 hectares of dense primary evergreen forest, fragmenting crucial wildlife corridors between the Anshi-Dandeli and Bhadra-Kudremukh landscapes. Experts warn of extreme landslide risks in this geologically fragile zone, where blasting for underground structures like tunnels and powerhouses with heavy industrial explosives threatens catastrophe. With an installed renewable energy capacity of 20.23 GW, Karnataka needs diversified solutions

such as Battery Energy Storage Systems, pumped storage, compressed air energy storage and gravity energy storage to manage the intermittency of renewable energy.

Critics acknowledge this need. But meeting this need by destroying forever, a carbon-storing ecological backbone such as the Sharavathi Lion-Tailed Macaque Wildlife Sanctuary, which justly enjoys the highest levels of legal protection, is indefensible.

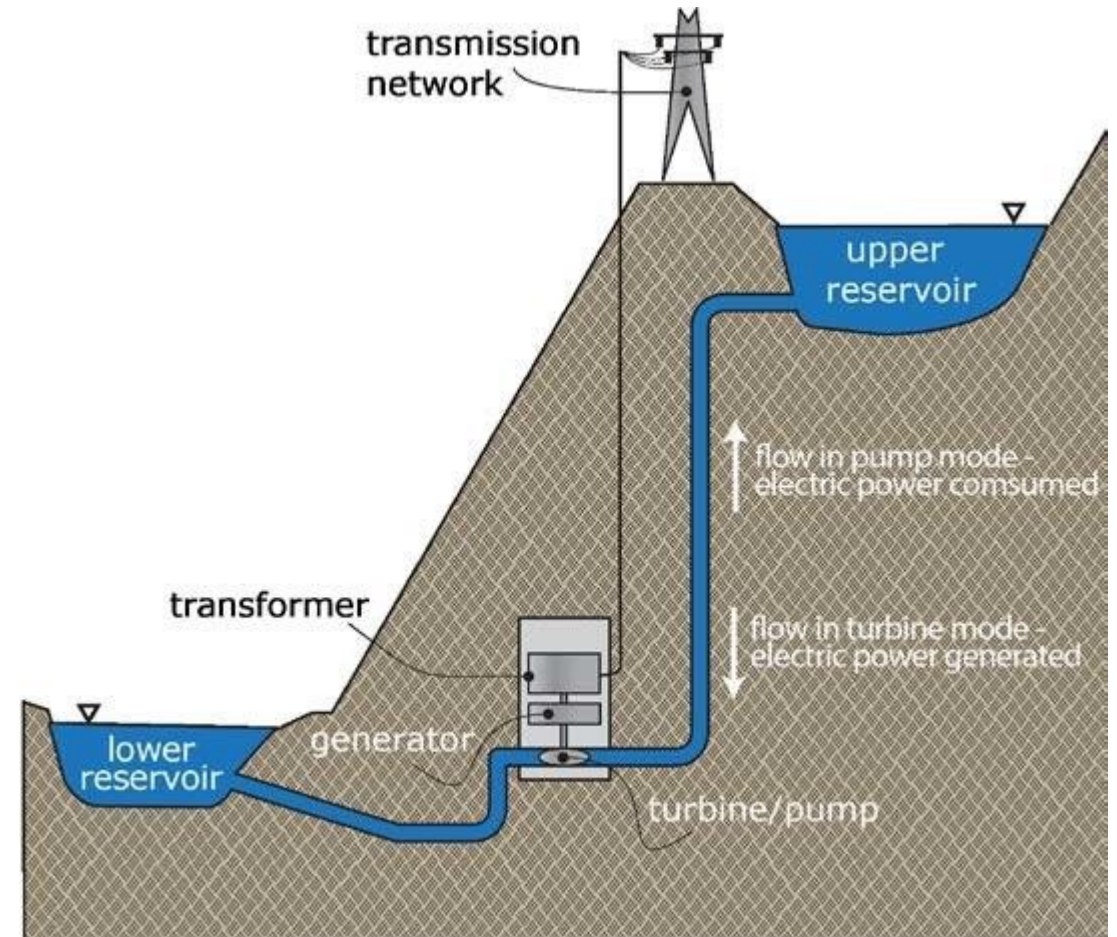
Cultural, ecological renewal

To defend the ecology of the irreplaceable Western Ghats entails renewing our ties to the natural world. No trite nostrum along the lines of “sustainable development” will suffice. Sustainability is about what we can go on taking from nature without finding the courage to face our organised, well-disguised greed. Ecological renewal and reciprocity imply that we honour what our ancestors have bequeathed to us — a stable, healthy ecology. It means we creatively renew a culture long integrated with nature, until the winds of accelerating industrial modernity begin to sunder them. We pass on what was passed down to us. We act for future generations the way our ancestors acted towards us. We act towards trees and animals, mountains and forests, rivers and coastlines with the reverence the cosmos deserves. This means putting an end to the plunder that corporate-consumer civilisation is routinely engaged in through mining and infrastructure, roads and expressways, dams, airports, and power plants. Not allowing *aranya* to be sacrificed to ecologically subsidised cities.

Shrivastava is a writer and teacher

Let the Western Ghats Breathe

- PSH is a type of hydroelectric energy storage.
- PSH is a fundamentally simple system that consists of two water reservoirs at different elevations.
- Working:
- When there is excess electricity available, such as during off-peak hours or from renewable sources like solar and wind, it is used to pump water from the lower reservoir to the upper reservoir.
- When there is a demand for electricity, the water is released from the upper reservoir back down to the lower reservoir, passing through turbines that generate electricity.
- The system also requires power as it pumps water back into the upper reservoir (recharge).



Let the Western Ghats Breathe

UPSC

Prelims

2024

Science and Tech

Q. Recently, the term “pumped-storage hydropower” is actually and appropriately discussed in the context of which one of the following?

- a** Irrigation of terraced crop fields
- b** Lift irrigation of cereal crops
- c** Long-duration energy storage
- d** Rainwater harvesting system

Thank You